3. Develop a C program using fork() that creates a parent and child process. Display the Process ID (PID), Parent Process ID (PPID), and process states at different stages of execution.

Design an experiment to observe process state transitions (Ready, Running, Waiting, Terminated) using Linux monitoring tools such as ps, top, and /proc. Document the observations.

OUTPUT :

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid;

    printf("Parent process started.\n");
    printf("Parent PID  : %d\n", getpid());
    printf("Parent PPID : %d\n", getppid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {

        printf("\n--- CHILD PROCESS ---\n");
        printf("Child PID  : %d\n", getpid());
        printf("Child PPID : %d\n", getppid());

        printf("Child is running...\n");

        sleep(10);

        printf("Child is terminating...\n");
        exit(0);
    }
    else
    {
        printf("\n--- PARENT PROCESS ---\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID  : %d\n", pid);

        printf("Parent is waiting for child...\n");

        wait(NULL);

        printf("Child has terminated.\n");
        printf("Parent continues execution.\n");
    }
```

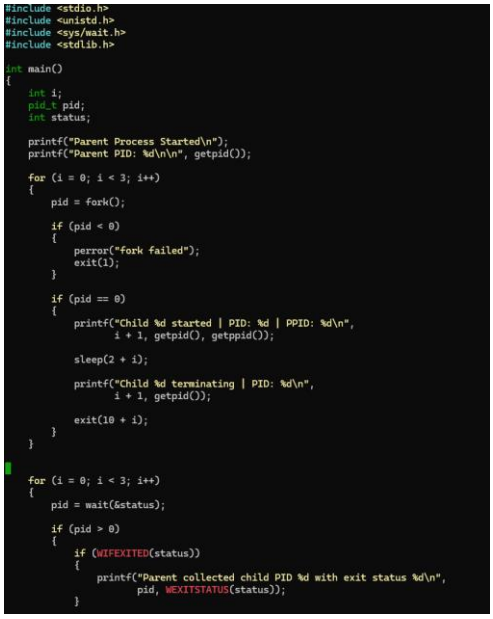
:

```
bhargav08@Bhargava:~/OS$ nano practical2.c
bhargav08@Bhargava:~/OS$ nano practical3.c
bhargav08@Bhargava:~/OS$ gcc practical3.c -o practical3
bhargav08@Bhargava:~/OS$ ./practical3
Parent process started.
Parent PID  : 554
Parent PPID : 318

--- PARENT PROCESS ---
Parent PID : 554
Child PID  : 555
Parent is waiting for child...

--- CHILD PROCESS ---
Child PID  : 555
Child PPID : 554
Child is running...
Child is terminating...
Child has terminated.
Parent continues execution.
```

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.

Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

OUTPUT :



```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    int i;
    pid_t pid;
    int status;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n\n", getpid());

    for (i = 0; i < 3; i++)
    {
        pid = fork();

        if (pid < 0)
        {
            perror("fork failed");
            exit(1);
        }

        if (pid == 0)
        {
            printf("Child %d started | PID: %d | PPID: %d\n",
                   i + 1, getpid(), getppid());

            sleep(2 + i);

            printf("Child %d terminating | PID: %d\n",
                   i + 1, getpid());

            exit(10 + i);
        }
    }

    for (i = 0; i < 3; i++)
    {
        pid = wait(&status);

        if (pid > 0)
        {
            if (WIFEXITED(status))
            {
                printf("Parent collected child PID %d with exit status %d\n",
                       pid, WEXITSTATUS(status));
            }
```

```
bhargav08@Bhargava:~/OS$ nano practical4.c
bhargav08@Bhargava:~/OS$ gcc practical4.c -o practical4
bhargav08@Bhargava:~/OS$ ./practical4
Parent Process Started
Parent PID: 655

Child 1 started | PID: 656 | PPID: 655
Child 2 started | PID: 657 | PPID: 655
Child 3 started | PID: 658 | PPID: 655
Child 1 terminating | PID: 656
Parent collected child PID 656 with exit status 10
Child 2 terminating | PID: 657
Parent collected child PID 657 with exit status 11
Child 3 terminating | PID: 658
Parent collected child PID 658 with exit status 12

All children have been collected.
Parent process terminating.
bhargav08@Bhargava:~/OS$
```